**THE UNITED REPUBLIC OF TANZANIA**
**PRESIDENT'S OFFICE**
**REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT**
**LONGIDO DISTRICT JOINT EXAMINATION**

**FORM FOUR MIDTERM EXAMINATION**
**FORM FOUR MOCK EXAMINATION**
**031 PHYSICS 1**

**Time: 3 Hours** **Tuesday, 8th April 2025 p.m.**

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## INSTRUCTIONS

1. This paper consists of sections **A, B** and **C** with total of eleven (11) questions.
2. Answer all questions from section **A** and **B** and any two (2) questions from section **C**.
3. Mathematical table and non –programmable calculator may be used.
4. Where necessary the following constants may be used
   - Acceleration due to gravity = 10 $ms^{-2}$
   - Density of water = 1.0 $gcm^{-3}$ or 1000 $kgm^{-3}$

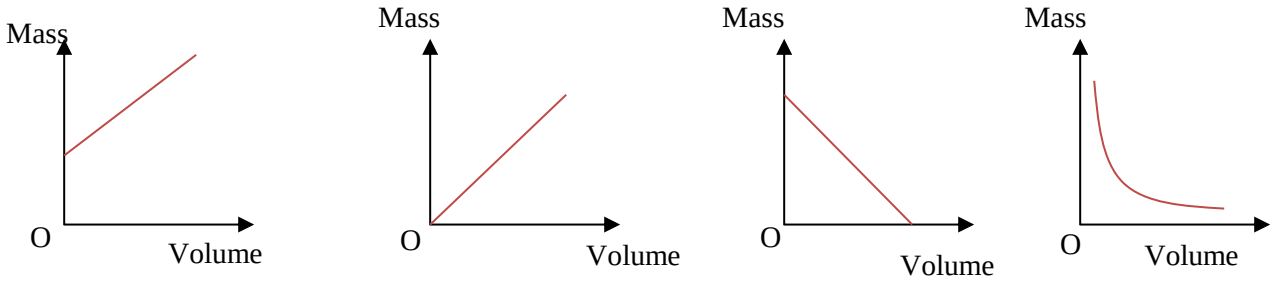
## SECTION A (16 Marks)
**Answer all questions**

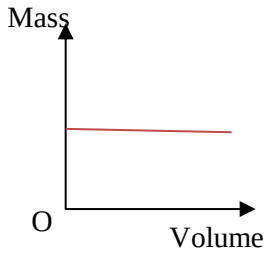
1. For each of the items (i) – (x), choose the correct answer from the given alternatives and write its letter beside the item number in the booklet provided. **(10marks)**

(i) How does studying physics contribute to our understanding of the natural world and the world and the universe?

A. Physics is not related to the study of the natural world and the universe
B. Physics helps us to understand the behavior of matter and energy, from the subatomic particles to celestial bodies
C. Physics only deals with artificial and human made systems.
D. Physics is not essential for understanding the natural world and the universe
E. Physics is very different from other sciences

(ii) A group of form four students from Seypalm Secondary School conducted an experiment to determine the density of wood. They measure the masses and volumes of different sized samples of wood. Furthermore, they plotted the graph based on the data collected as shown in the figure below. Which graph depicts their results?






E

(iii) The balls in figure below have the same size. Which one will hit the ground first if they are dropped at the same time and the same height?



Wooden ball Wooden ball Wooden ball

A. Metal first, then wood, then plastic last
B. Wood first, then plastic, then metal last.
C. They will all hit the ground at the same time
D. There is no way to tell
E. Plastic first, then the rest

(iv) Colours are produced when light passes through a glass prism because:
A. Light waves interfere
B. Glass prism colours the light
C. In glass different colours travel at different speed
D. Different colours are filtered
E. Diffraction of light occurs

(v) Seypalm wrapped himself with a few layers of cloth on a cold winter night. This helped himself to keep warm because
A. Friction between the layers of cloth produce heat
B. Air trapped between the layers of cloth is a bad conductor of heat
C. Cloth is at higher temperature than the temperature surrounding
D. Cloth is a good conductor of heat
E. Clothes are less expensive, that is why he could afford several layers of them.

(vi) In their endeavor to harness renewable energy for use, scientists have now invented 'biodiesel'. This is the production of cheap and reliable diesel from seeds of castor oil and croton trees. Which of the following likely to frustrate their efforts?
A. Lack of market
B. Due to climatic patterns, the supply of seeds may be inadequate
C. Unscrupulous traders
D. It is expensive to set up the plants
E. Clean energy from the biodiesel

(vii) When will you find field lines stronger?
A. When the field lines are not seen
B. When the field lines are crowded
C. When the lines are together
D. When the lines are far apart
E. When the magnets are weak

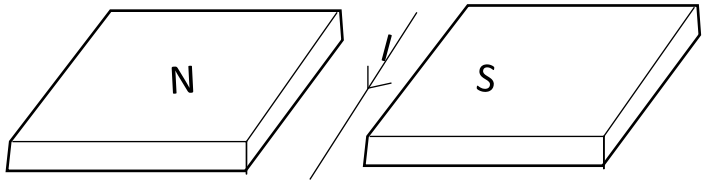
(viii) Why a large ship floats while a small metal ball sinks?

A. The density of the ship is greater than the density of water
B. The density of the ship is smaller than the density of water
C. The mass of water displaced by the ship is greater than the weight of the ship
D. The mass of the small metal ball is greater than the mass of the large ship.
E. The density of the metal ball is smaller than the density of water

(ix) Select the process which is involved in producing reverberation

A. Refraction
B. Multiple reflection
C. Interference
D. Reflection
E. Diffraction

(x) A wire carries a current 'I' horizontally between the magnetic poles N and S which face each other on a table (see the figure below)




The direction of the force on the wire due to the magnet is:

A. From N to S
B. From S to N
C. Opposite to the current direction
D. Vertically upwards
E. In the direction of the current

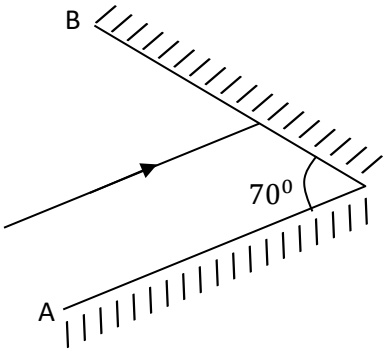
2. Match the items in list A with responses in list B by writing the letter of the correct response beside. **(6marks)**

| **LIST A** | **LIST B** |
|---|---|
| i. Mist<br>ii. Snow<br>iii. Frost<br>iv. Dew<br>v. Fog<br>vi. Hail stone | A. Is formed when the dew point is below the freezing point ($^0$C)<br>B. Phenomenon of melting under pressure and refreezing when the pressure is reduced<br>C. Is water in the form of droplets that appears on exposed objects in the morning or in the evening due to condensation<br>D. Process by which a liquid turns into a vapour when it is heated to its boiling point<br>E. Is a cloud floating just above the ground. It reduces visibility to less than 200m<br>F. Are water droplets in clouds formed due to super cooling below $^0$C without freezing<br>G. Is a deposit of small white ice crystals formed on the ground or other surfaces when the temperature falls below freezing point<br>H. Is the condensation of vapour into water droplets occurring near the ground |

## SECTION B (54 Marks)

**Answer all questions in this section**

3. (a) The figure below shows two plane mirrors, A and B inclined at $70^0$to each other. A ray of light parallel to mirror A is inclined to mirror B. What is the angle of reflection of the light ray on mirror A? **(4marks)**

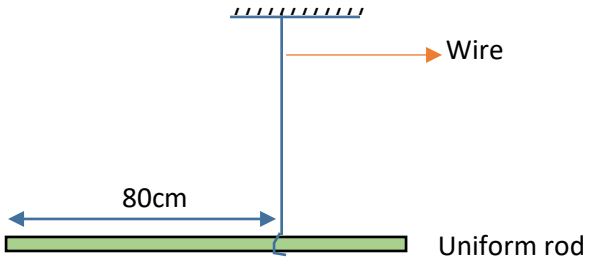



(b) A projector lens is used to produce a sharp image of an object when the distance between the object and screen is 160 cm. If the linear magnification is 7, calculate the focal length of the lens. **(5marks)**

4. (a) (i) A hole at the bottom of a ship is more dangerous than one near the surface. Explain why? **(2marks)**

(ii) Briefly explain, why heavy trucks are made up with many tyres? **(2marks)**

(b) A uniform rod with a mass of 120g and length 130cm is suspended by a wire from a point 80 cm from the rods left end.



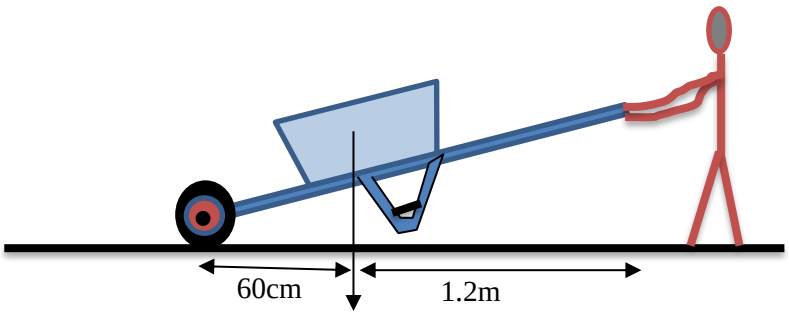

(i) What mass must be hang from the right end of rod for it to be in equilibrium? **(3marks)**

(ii) What will be the tension in the wire? **(2marks)**

5. (a) (i) Apply Newton's first law of motion to explain why an object pulled along the ground with a constant velocity has zero net force but force on it is not zero **(2 marks)**

(ii) Suppose an object is not moving, can you conclude that there are no forces acting on it? Explain your answer. **(2 marks)**

(b) A wheelbarrow is to be used to move a stone weighing 600N. The wheelbarrow weighs 100N. The weight of wheelbarrow and its content act at a point on wheelbarrow as shown in figure below.




Ignoring frictional forces, determine

(i) The mechanical advantage of wheelbarrow **(3marks)**

(ii) The minimum amount of effort needed to just lift the legs of wheel barrow off the ground **(2marks)**

6. (a) A beaker containing water ice is heated from $-5^0$C to $0^0$C and then from $0^0$C to $15^0$C. With the aid of a diagram, explain the variation of density with temperature. **(4 marks)**

(b) A heater of 500w is used to raise the temperature of 50kg of material of specific heat capacity of 960J/kgK, from $18^0$C to $38^0$C. Assume that all the heat from the heater is given to the material. Calculate ;

(i) Heat capacity of the material **(2 marks)**

(ii) The time taken in seconds **(3 marks)**

7. (a) (i) The half-life of a certain radioactive substance is 64 days. Explain the meaning of this statement**. (2 marks)**

(ii) The half-life of $60_{Co}$ is 5 years. How much time does it take for $(\frac{3}{4})^{th}$ of its initial mass to disintegrate? **(4 marks)**

(b) Thorium disintegrate in the following manners:

$90^{Th^{232}} .\rightarrow 88^{Ra^{228}} \rightarrow 89^{Ac^{228}} .\rightarrow 90^{Th^{228}} \rightarrow 88^{Ra^{224}}$. State the particle emitted at each part of the disintegration. **(4 marks)**

8. (a) Why a ship rises a bit when it enters in a saline sea from a river? **(3 marks)**

(b) A mass of sealed container of volume $400cm^3$ is 48g.What is the mass of steel of density $8000kg/m^3$ that can be attached beneath the container, if the container is to float when submerged in fresh water? **(6 marks)**

## SECTION C (30 Marks)
## Answer only two (2) questions from this section

9. (a) (i) With support of a diagram, what will be the effect on electron as it passes between two plates A and B, one of which carrying a + ve charge and the other carrying a –ve charge? Remember electron carries negative charge **(3 marks)**

(ii) Explain why it is more dangerous to walk barefooted during lightning storms? **(3 marks)**

(b) If the resistance is doubled and the potential difference across it is reduced to one third, what is the ratio of old current to new current? **(4marks)**

(c) Each of the two new dry cells has an e.m.f of 1.5 volts and internal resistance of 1.0 ohm. The two cells are connected in parallel and the combination connected to a 10-ohm resistor.

Find the current and heat developed per second on the 10-ohm resistor. **(5 marks)**

10. (a) In which part of a fridge or a microwave oven are magnetic strips installed? Why? **(5 marks)**

(b) A helicopter is hovering at an altitude of 200m above the surface of a lake. A speaker on the helicopter is sending out sound waves, which are reflected from both the surface of water and bottom of the lake. If the difference in arrival times of the two echoes is measured to be 0.24seconds, what is the depth of the lake? (The atmospheric temperature is $20^0$C). **(6 marks)**

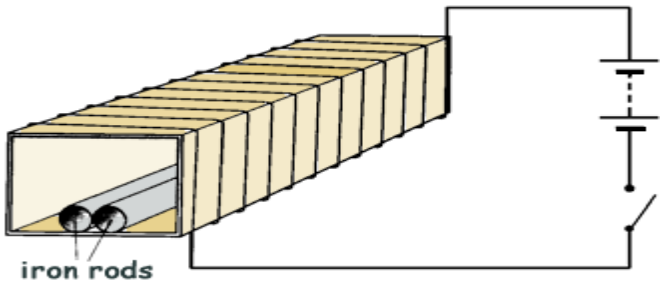
(c) As waves pass by a duck floating on a lake or a pond, the duck bobs up and down but remains in one place. Explain whythe duck is not carried along by the wave motion? **(4 marks)**

11. (a) (i) What is meant by the term 'solenoid electromagnetic inductor?' **(3 marks)**

(ii) A transformer used to step down 240V mains supply to 12V for a laboratory use. If the primary coil has 600 turns; determine the turns in the secondary coil**. (3 marks)**

(b) The diagram shows a wire wrapped around a box to form a rectangular coil and circuit. The coil has two iron rods in it. The rods are parallel and touching. They are free to move.

Diagram




When the switch is closed, the two rods move apart

i. Explain why this happens **(3mark)**

ii. The switch is then opened to break the circuit. What, if anything , happens to the two iron rods **(3mark)**

iii. One of the two iron rod is taken out and replaced with a brass rod. The rods are parallel and touching, what, if anything, will happen to the rods the when switch is closed? Explain your answer. **(3mark)**